- **Project Sentinel: Enterprise-Scale SOC Simulation & Adversary Emulation**
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- **Role:** Junior SOC Analyst / Cybersecurity Researcher
- **Status:** Completed & Documented
- **Contact:** Contact: Check My Cybersecurity Portfolio
- **Executive Summary**
- In the modern threat landscape, **Visibility** is the only antidote to **Obscurity**. This project is a rigorous technical demonstration of an end-to-end Security Operations Center (SOC) lifecycle. It bridges the gap between offensive Red Team tactics and defensive Blue Team detection engineering.
- The primary mission was to build a distributed environment that captures authentic system telemetry, allowing for the analysis of the **Cyber Kill Chain**—from the first "knock" on the digital door to the final unauthorized access and post-exploitation monitoring.
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- **Technical Infrastructure & Network Topology**
- Unlike standard virtualized labs, this project utilizes a **Distributed Physical-Virtual Hybrid Model**. This setup ensures that network artifacts (latency, TTL, packet fragmentation) are realistic and reflect genuine enterprise environments.
  - **Node Alpha: The Command & Control (C2) - Attacker**
- **Operating System:** Kali Linux 2024.x (Physical Host).
- **Network Role:** Remote External Adversary.
- **Tooling Suite:** * **Enumeration:** Nmap (Network Mapper) with custom NSE scripts.
  - **Exploitation:** Metasploit Framework (MSF) for modular vulnerability validation.
  - **Credential Access:** Hydra (Parallelized login cracker).
  - **Node Beta: The Defensive Sentinel - SIEM Manager**
- **Platform:** Wazuh SIEM & XDR (Open Source).
- **Architecture:** * **Wazuh Indexer:** For high-speed security event indexing and storage.
  - **Wazuh Manager:** Distributed log processing and rule-matching engine.
  - **Wazuh Dashboard:** Real-time data visualization and incident triage interface.
- **Network Configuration:** Static IP assignment (**192.168.1.3**) to ensure persistent agent-manager heartbeats.
  - **Node Gamma: The Hardened Target - Endpoint**
- **Operating System:** Windows 10 Pro (Target Instance).
- **Security Integration:**
  - **Wazuh Agent:** Deployed as a service for real-time log shipping.
  - **Sysmon (System Monitor):** Deployed with a granular XML configuration to monitor sub-process injection and network connections.

- **Log Enrichment:** Configured to capture **Event ID 4624 (Success)** and **4625 (Failure)** with full metadata.

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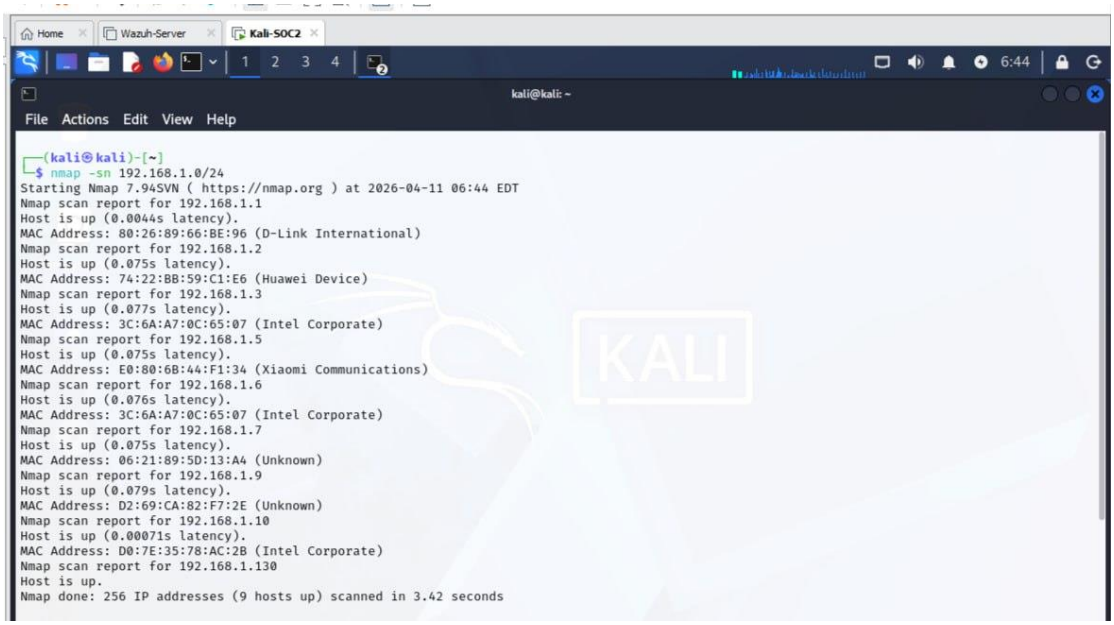
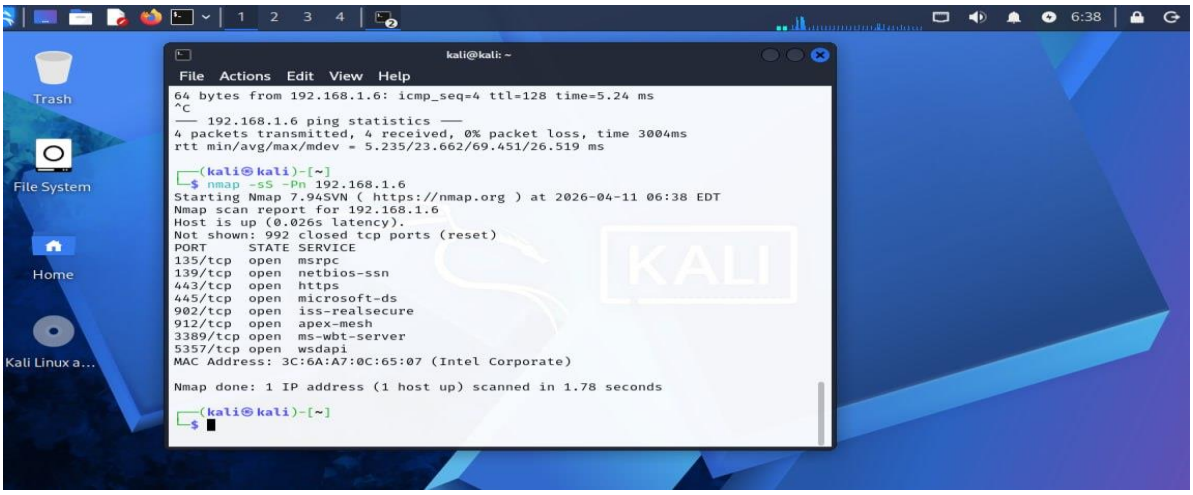
- **Phase I: Advanced Reconnaissance & Surface Mapping**
  - **Macro-Level Discovery (Host Identification)**
- I initiated a non-intrusive ARP-based discovery to identify active assets within the subnet.
- **Command:** nmap -sn 192.168.1.0/24
- **Analysis:** This reduced the scope to **9 active nodes**, identifying the target at 192.168.1.6.



  - **Micro-Level Enumeration (Deep Fingerprinting)**
- A multi-vector scan was executed to determine service versions and OS build details.
- **Command:** nmap -sS -sV -p 1-65535 -T4 -A -Pn 192.168.1.6
- **Intelligence Reports:**
  - **Service:** Microsoft RDP active on **3389**.
  - **Service:** Microsoft SMB active on **445**.
  - **Fingerprint:** Detected **VMware Workstation** infrastructure via service banners.

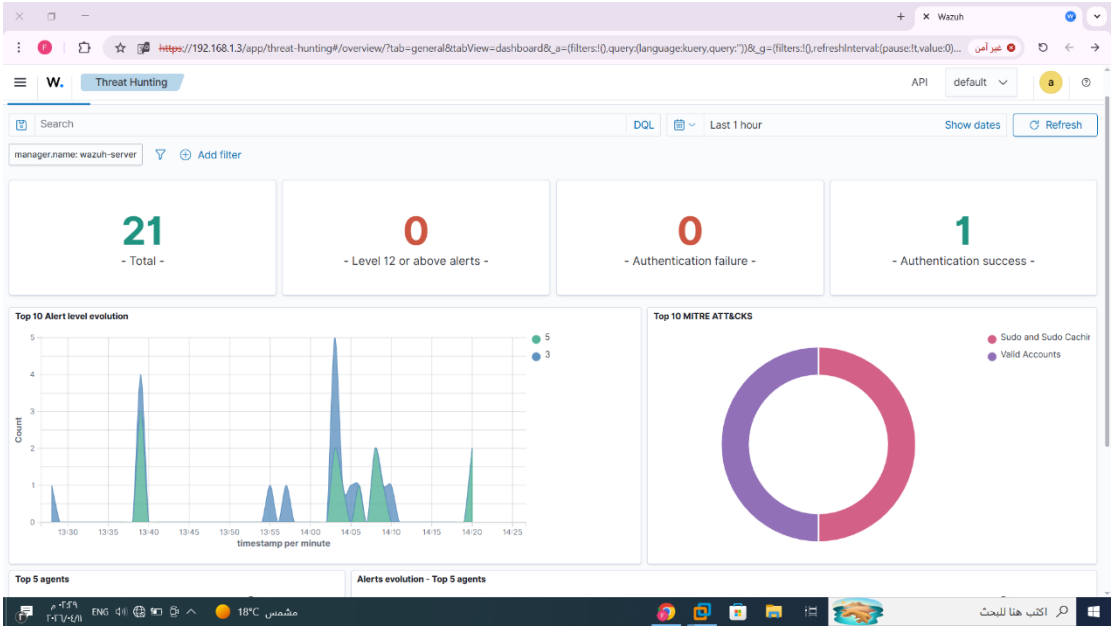


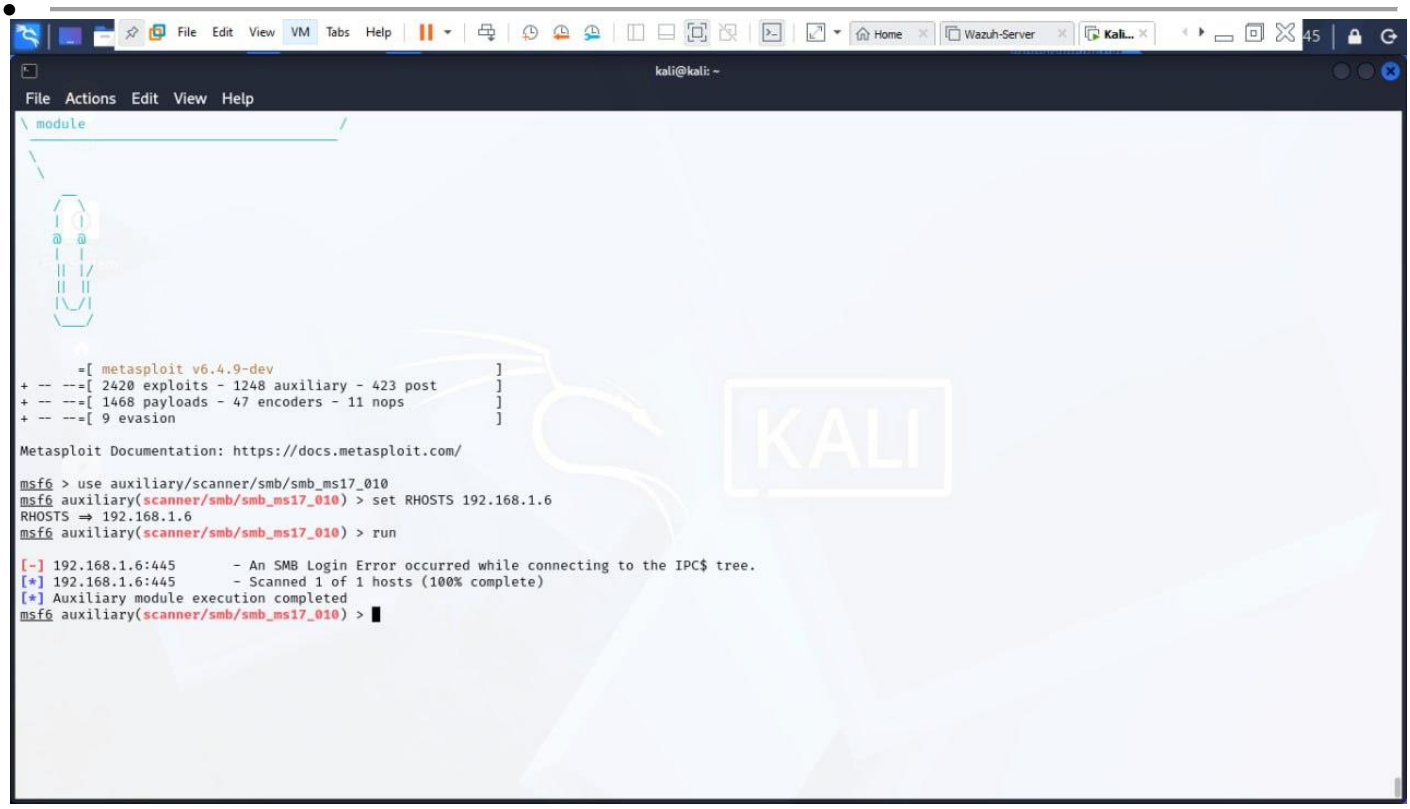
```
(kali㉿kali)-[~]
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-11 06:44 EDT
Nmap scan report for 192.168.1.1
Host is up (0.0044s latency).
MAC Address: 80:26:89:66:BE:96 (D-Link International)
Nmap scan report for 192.168.1.2
Host is up (0.075s latency).
MAC Address: 74:22:BB:59:C1:E6 (Huawei Device)
Nmap scan report for 192.168.1.3
Host is up (0.077s latency).
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)
Nmap scan report for 192.168.1.5
Host is up (0.075s latency).
MAC Address: E0:80:6B:44:F1:34 (Xiaomi Communications)
Nmap scan report for 192.168.1.6
Host is up (0.076s latency).
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)
Nmap scan report for 192.168.1.7
Host is up (0.075s latency).
MAC Address: 06:21:89:5D:13:A4 (Unknown)
Nmap scan report for 192.168.1.9
Host is up (0.079s latency).
MAC Address: D2:69:CA:82:F7:2E (Unknown)
Nmap scan report for 192.168.1.10
Host is up (0.00071s latency).
MAC Address: D0:7E:35:78:AC:2B (Intel Corporate)
Nmap scan report for 192.168.1.130
Host is up.
Nmap done: 256 IP addresses (9 hosts up) scanned in 3.42 seconds
```

```
MAC Address: 3C:6A:A7:0C:65:07 (Intel Corporate)
No exact OS matches for host (If you know what OS is running on it, see https
://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.94SVN%E=4%D=4/11%OT=135%CT=1%CU=33439%PV=Y%DS=1%DC=D%G=Y%M=3C6A
OS:A7%TM=69DA26DD%P=x86_64-pc-linux-gnu)SEQ(SP=101%GCD=1%ISR=106%TI=I%CI=I%
OS:TS=U)SEQ(SP=101%GCD=1%ISR=106%TI=I%CI=I%II=I%SS=S%TS=U)SEQ(SP=102%GCD=1%
OS:ISR=106%TI=I%CI=I%II=I%SS=S%TS=U)OPS(O1=M5B4NW8NNS%O2=M5B4NW8NNS%O3=M5B4
OS:NW8%O4=M5B4NW8NNS%O5=M5B4NW8NNS%O6=M5B4NNS)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W
OS:4=FFFF%W5=FFFF%W6=FF70)ECN(R=Y%DF=Y%T=80%W=FFFF%O=M5B4NW8NNS%CC=N%Q=)T1(
OS:R=Y%DF=Y%T=80%S=O%A=S+%F=AS%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=O%F=AR%O=
OS:%RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80%W
OS:=0%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T
OS:4(R=Y%DF=Y%T=80%W=0%S=O%A=O%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=O%
OS:F=AR%O=%RD=0%Q=)T5(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=
OS:Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=O%A=O%F=R%O=%RD
OS:=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=O%F=AR%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%
OS:S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPC
OS:K=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=80%CD=Z)

Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/o:vmware:Works
tation:15.5.5

Host script results:
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled but not required
|_nbstat: NetBIOS name: FATER, NetBIOS user: <unknown>, NetBIOS MAC: 3c:6a:a7
:0c:65:07 (Intel Corporate)
| smb2-time:
|   date: 2026-04-11T10:47:51
|_  start_date: N/A

TRACEROUTE
HOP RTT     ADDRESS
1   5.20 ms 192.168.1.6

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 165.08 seconds
```

Blue Team Detection (Early Warning)

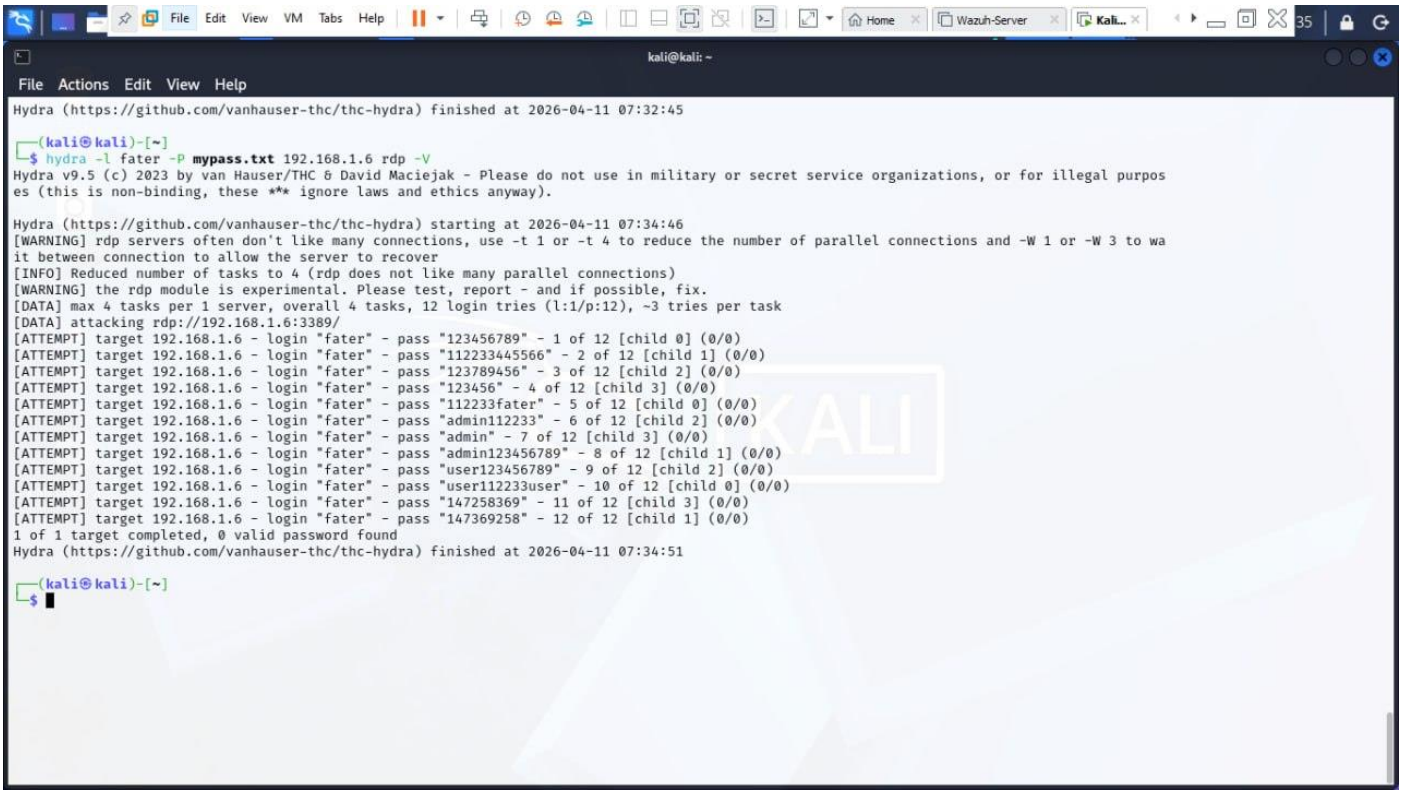
- The **Wazuh SIEM** immediately detected the "Port Scanning" activity.

- **Alert Triggered:** Nmap scanning detected / Host discovery.
- **Analyst Insight:** Early detection marks the transition from "Internet Noise" to "Targeted Interest".
- 

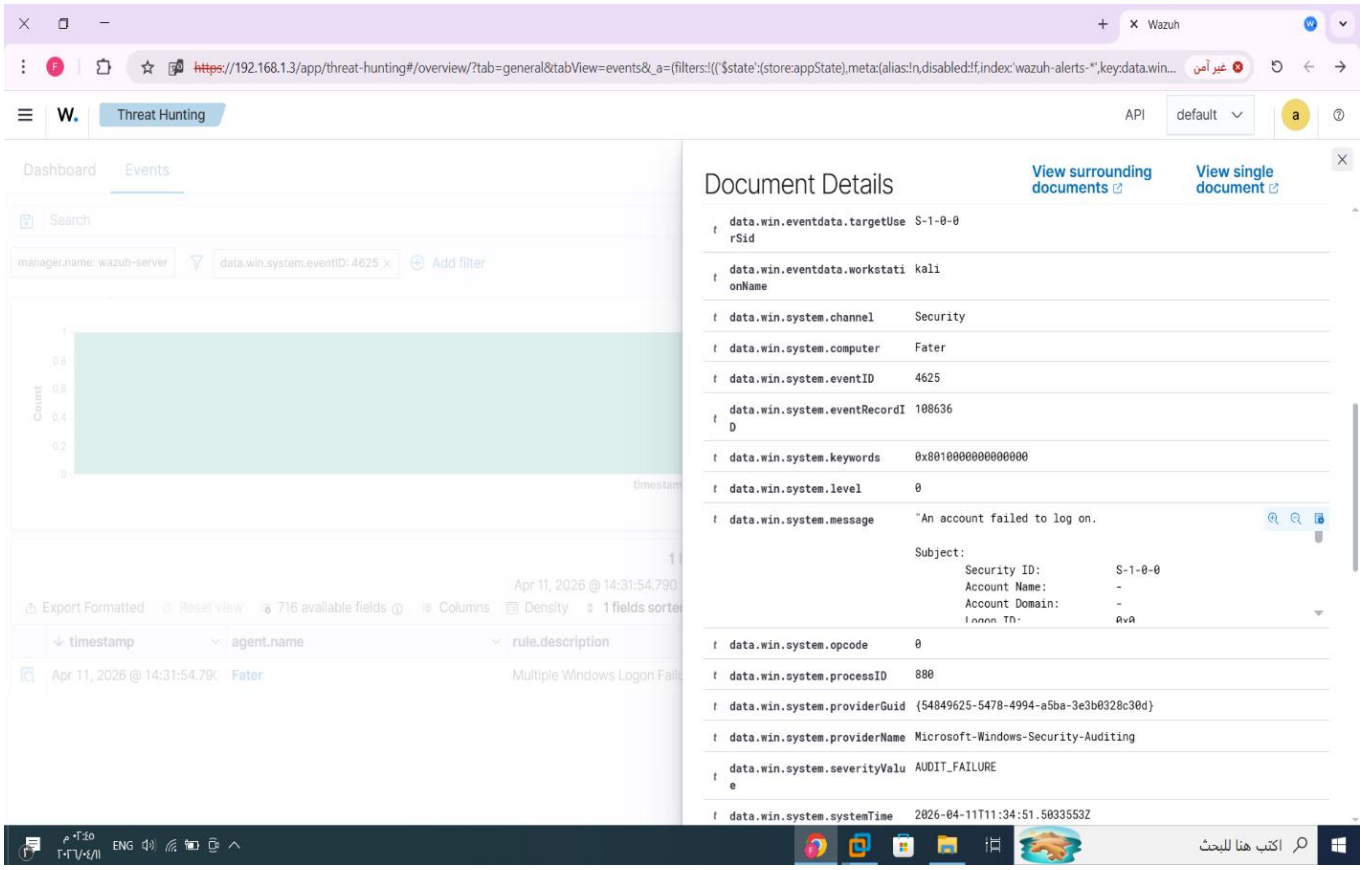
- **Phase II: Vulnerability Validation (The Metasploit Pivot)**
- Before launching noisy brute-force attacks, I conducted a "Silent Probe" using the **Metasploit Framework**.
- **Focus:** auxiliary/scanner/smb/smb_ms17_010.
- **Result:** The target was **Patched** against EternalBlue.
- **Strategic Shift:** The failure of unauthenticated RCE forced a shift to **T1110 (Brute Force)**—targeting Identity instead of Software Bugs.
- 

- **Phase III: Credential Access & Breach (Hydra)**
  - **The Brute-Force Offensive**
- Using **Hydra**, I targeted the RDP service with a dictionary-based attack.

- **Command:** hydra -l user -P /path/to/wordlist.txt 192.168.1.6 rdp

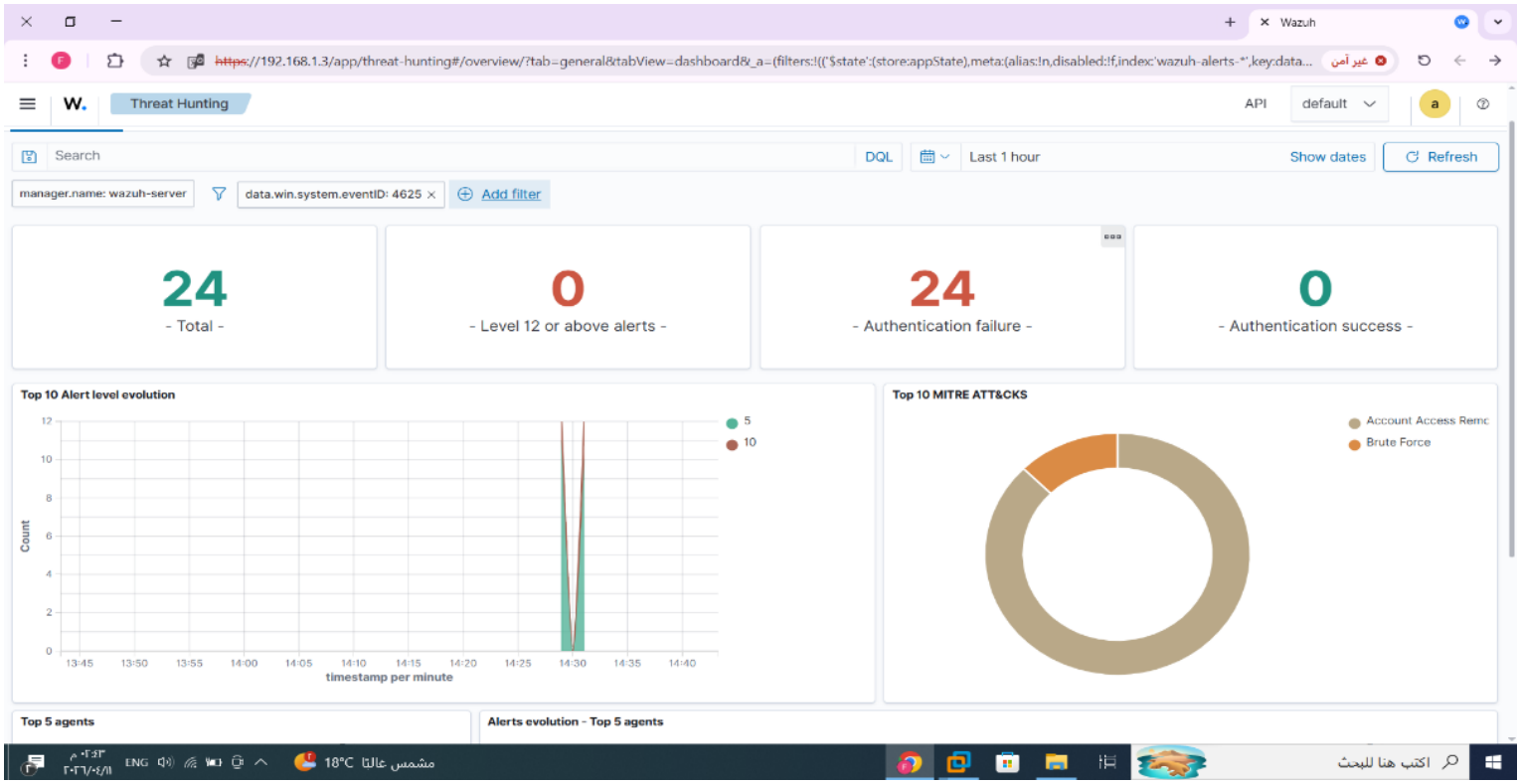


- **SOC Impact:** The SIEM recorded a massive spike in **Event ID 4625** (Logon Failure).

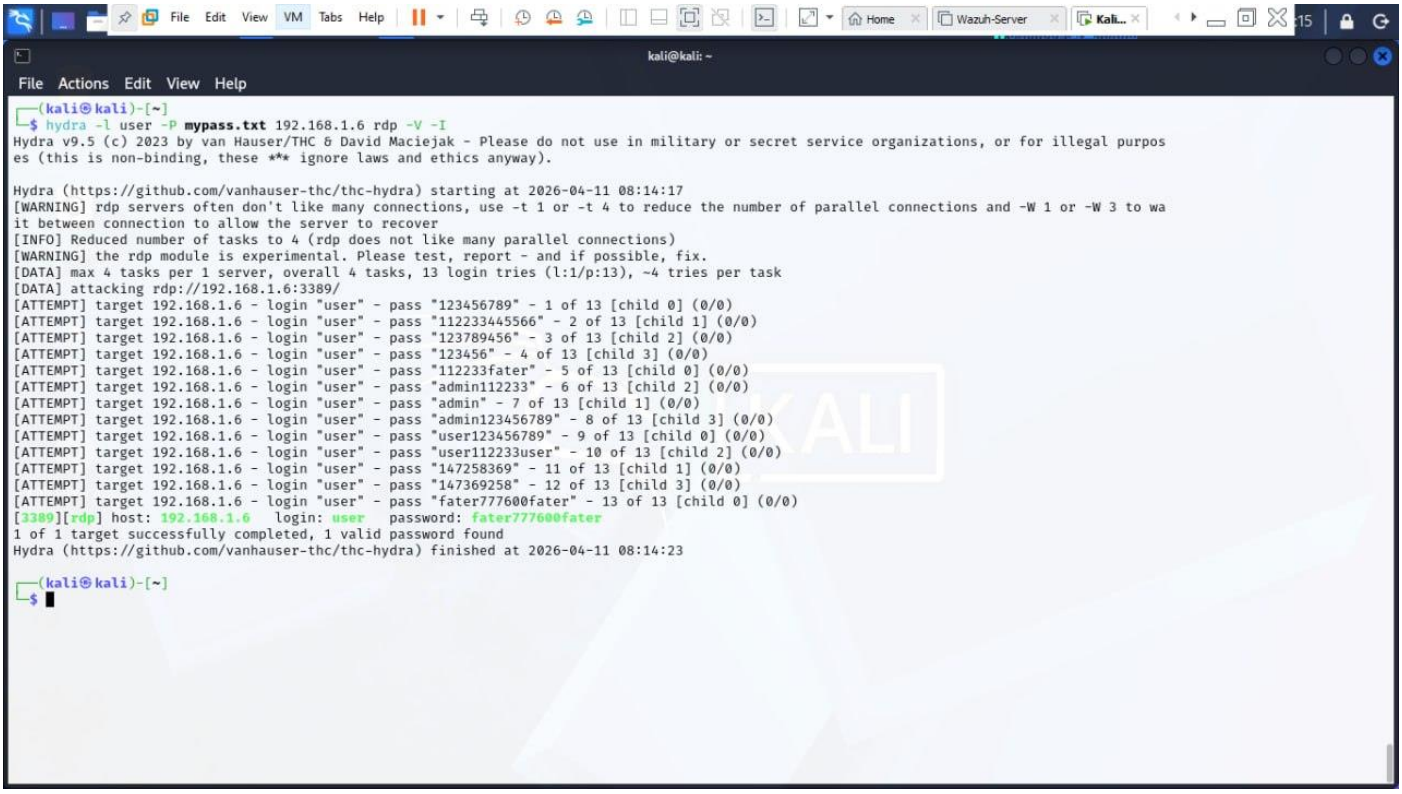


- **Authentication Correlation**

- Wazuh correlated **24 failed login attempts** within a 60-second window, triggering a **Level 10 (High Severity)** alert.

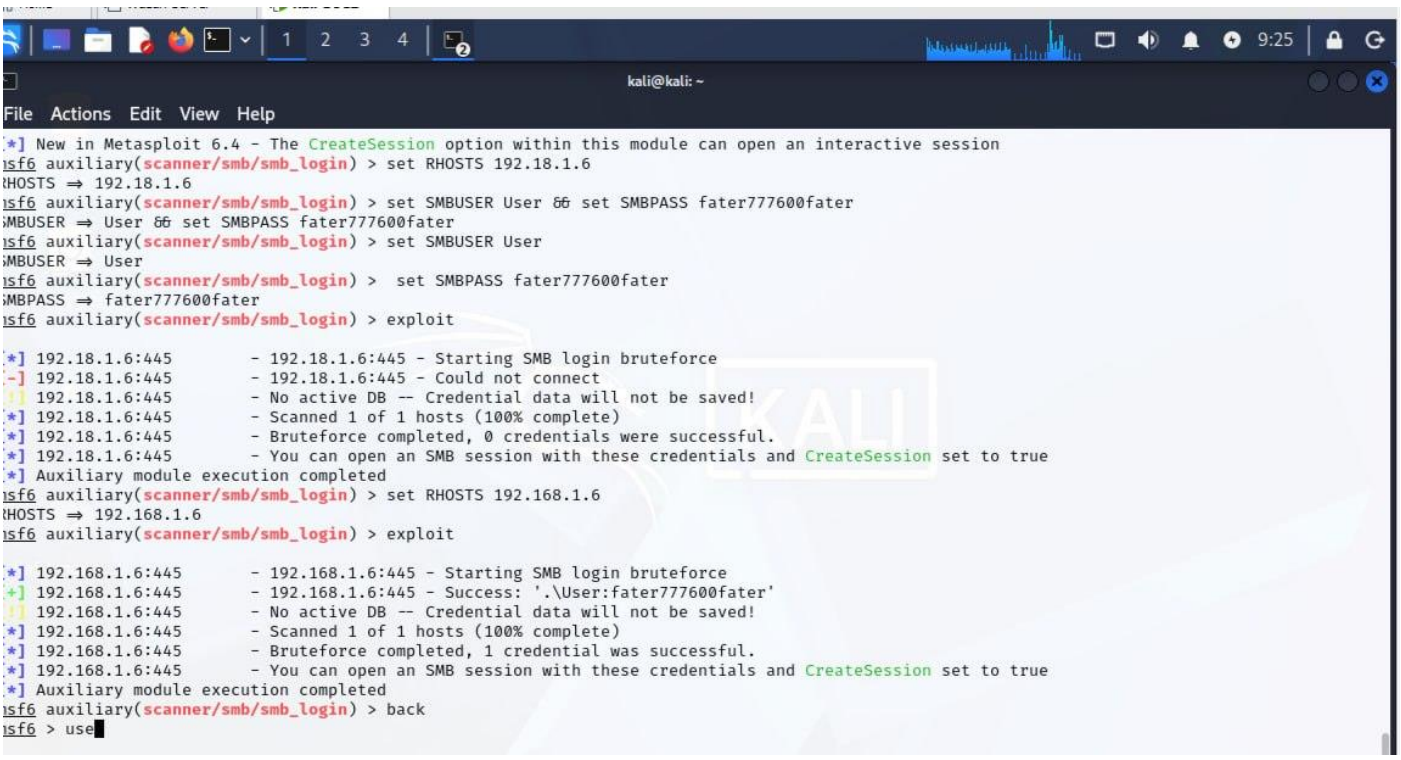


- The brute-force attack was successful. Hydra identified valid credentials for the RDP service.



- **Phase IV: Initial Access & Post-Exploitation Monitoring**
    - **Access Validation (Metasploit)**
- Credentials were confirmed via Metasploit's scanner/smb/smb_login module.

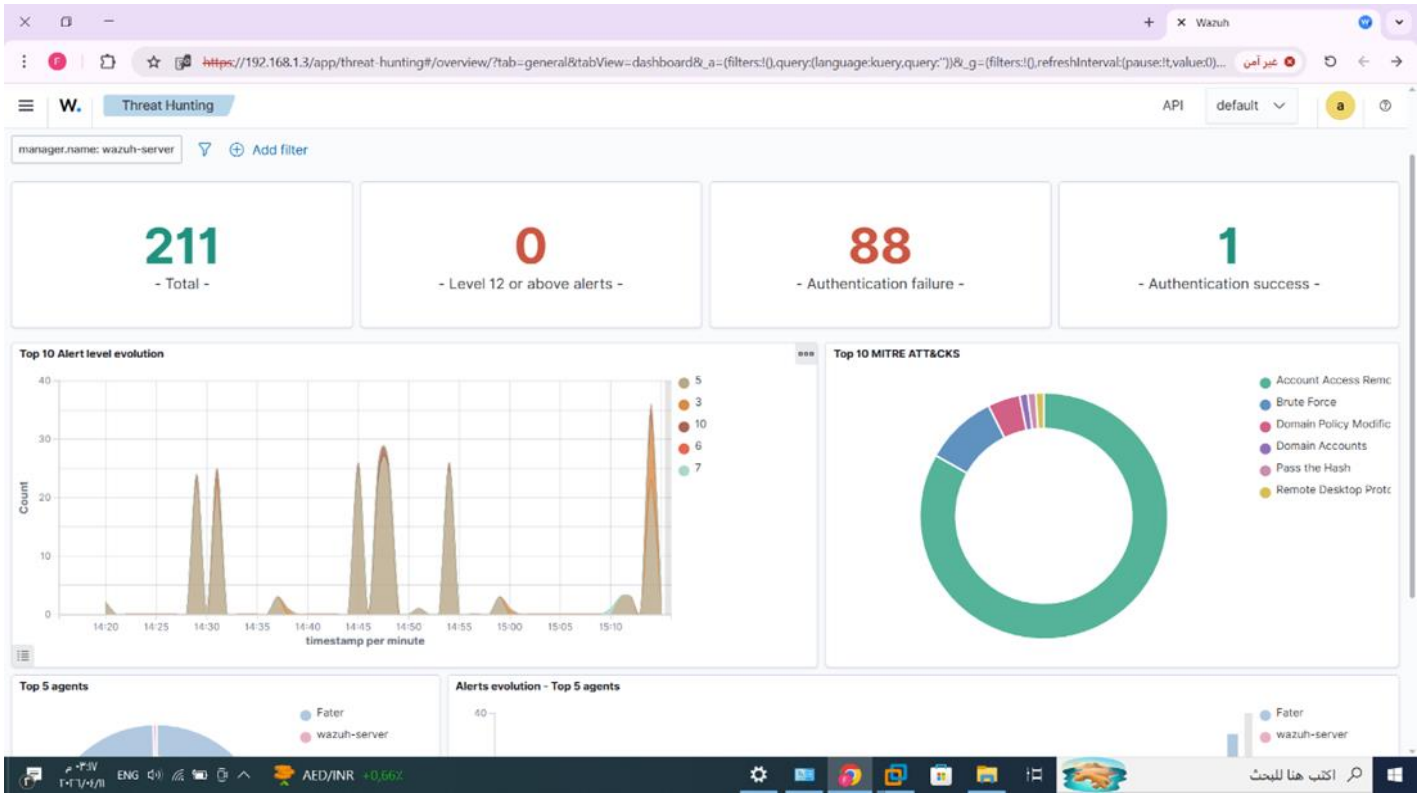
- **Outcome: SUCCESSFUL AUTHENTICATION**.

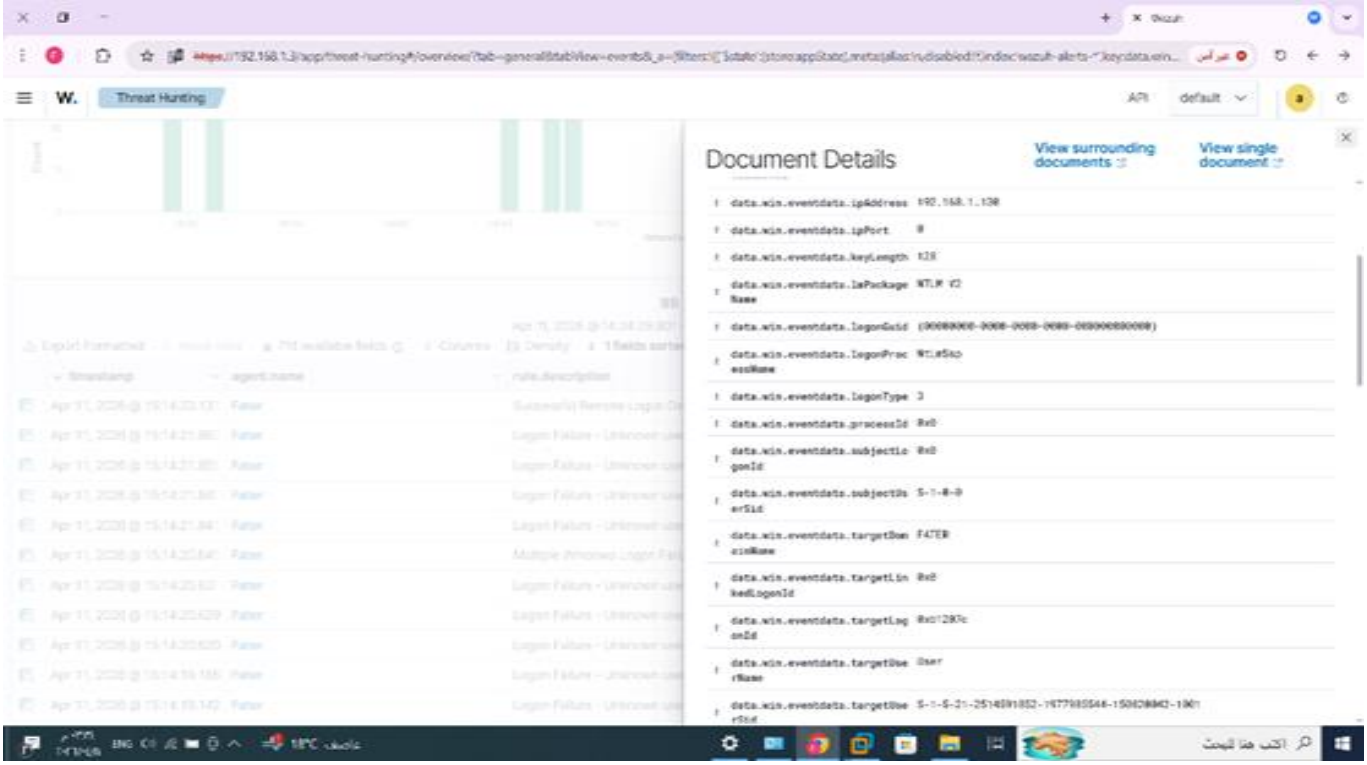


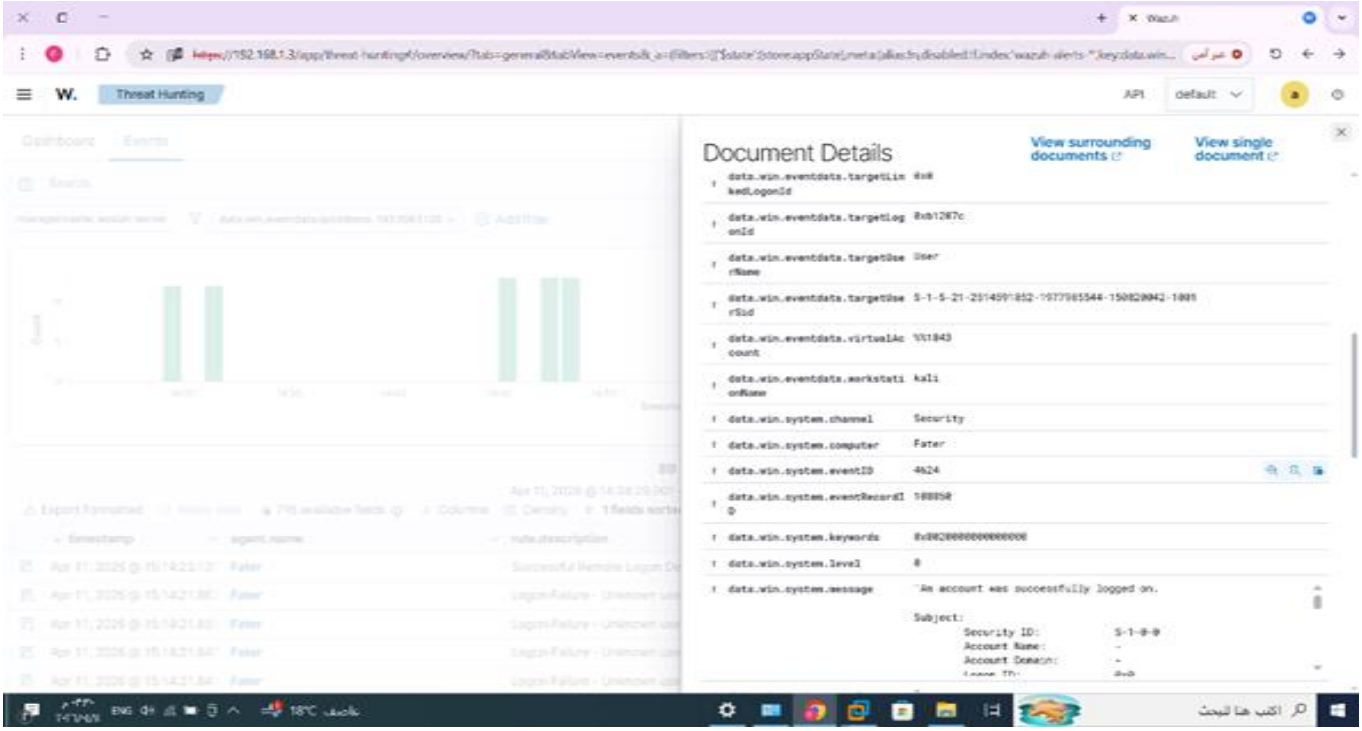
Detection of Successful Breach

- The SIEM immediately flagged the **Event ID 4624 (Logon Type 3)**.

- **Analyst Action:** I cross-referenced the successful logon time with the preceding 88 failures, confirming a successful breach.







MITRE ATT&CK Framework Mapping

• **Stage**	• **Tactic ID**	• **Technique**	• **Detection Source**
• **Reconnaissance**	• T1595	• Active Scanning	• Wazuh Network Events
• **Credential Access**	• T1110	• Brute Force	• Win Security Log (4625)
• **Initial Access**	• T1078	• Valid Accounts	• Win Security Log (4624)
• **Discovery**	• T1046	• Network Service Scanning	• Metasploit Auxiliary Logs

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- **Remediation & Hardening Roadmap**
- **Identity Security:** Enforce **Account Lockout Policies** (e.g., lock for 30 mins after 5 failures).
- **Attack Surface Reduction:** Isolate SMB and RDP behind a **VPN** or **RDP Gateway**.
- **Active Response:** Implemented a Wazuh script to automatically block any IP triggering a Port Scan or Brute-force alert for a 24-hour period.
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- **Contact & Support**
- Developed and Documented by **Fater Alrome && Ali Dalla**. For inquiries or professional collaboration, please reach out via Contact: Check My Cybersecurity Portfolio.